



CATALOGS OF NEBULOUS OBJECTS

Besides individual stars, various other types of object, such as star clusters, nebulae, and galaxies, have practically fixed positions on the celestial sphere. Most of these objects appear as no more than hazy blurs in the sky, even through a telescope. The first person to catalog such objects was a French astronomer, Charles Messier (see panel, below), in the 18th century. He compiled a list of 110 hazy objects, though none of these are from the southern polar skies—that is because Messier carried out his observations from Paris, and anything in declination below 40°S was below his horizon. In 1888, a much larger catalog called the *New General Catalog of Nebulae and Star Clusters* (NGC) was published, and this was later expanded by what is called the *Index Catalog* (IC). To this day, the NGC and IC are important catalogs of nebulae, star clusters, and galaxies. Their current versions cover the entire sky and provide data on more than 13,000 objects, all identified by NGC or IC numbers. In addition, several hundred specialized astronomical catalogs are in use, covering different types of objects, parts of the sky, and regions of the electromagnetic spectrum. Most catalogs are now maintained as computer databases accessible over the Internet.

NEW GENERAL CATALOG

More than 150 New General Catalog (NGC) objects lie within the constellation Ursa Major. Two are shown here, both spiral galaxies in a region around the Great Bear's forelegs, not far from Theta (θ) Ursae Majoris. NGC 2841 has delicate, tightly wound arms, within which astronomers have recorded many supernovae explosions. NGC 3079 has an active central region, from which rises a lumpy bubble of hot gas, 3,500 light-years wide, driven by star formation.



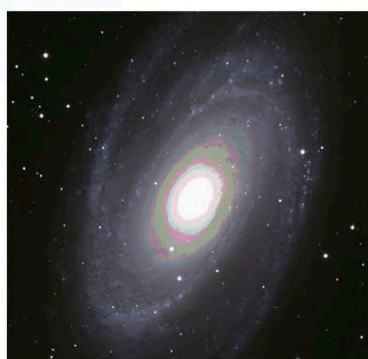
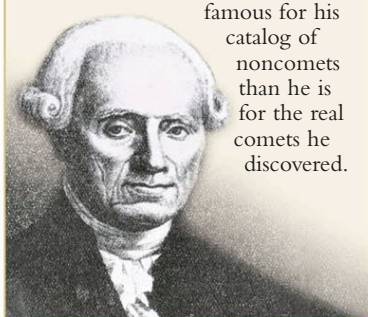
NGC 2841, A SPIRAL GALAXY



NGC 3079, A SPIRAL GALAXY VIEWED EDGE-ON

CHARLES MESSIER

The French comet-hunter Charles Messier (1730–1817) compiled a catalog of 110 nebulous-looking objects in the sky that could be mistaken for comets. Not all of them were discovered by himself—many were spotted by another Frenchman, Pierre Méchain, and yet others had been found years earlier by astronomers such as Edmond Halley. Messier's first true discovery was M3, a globular star cluster in Canes Venatici. Ironically, Messier is more famous for his catalog of noncomets than he is for the real comets he discovered.



M81, A SPIRAL GALAXY (SEE P.302)



M82, AN IRREGULAR GALAXY (SEE P.304)



M97, A PLANETARY NEBULA



M108, A SPIRAL GALAXY

THE MESSIER CATALOG

Messier's catalog includes 57 star clusters, 40 galaxies, 1 supernova remnant (the Crab Nebula), 4 planetary nebulas, 7 diffuse nebulas, and 1 double star. Of these Messier objects, 8 lie in the constellation of Ursa Major, of which 5 are shown here. Each is denoted by the latter M followed by a number. The planetary nebula M97 is also called the Owl Nebula. Galaxies M81 and M82 are neighbors in the sky and can be viewed simultaneously with a good pair of binoculars. M109 lies close to the star Phecda—Gamma (γ) Ursae Majoris—in the Big Dipper.



M109, A BARRED SPIRAL GALAXY